Portfolio Review Memo: Default Balanced Wealth Portfolio

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Executive summary

Recommendation: conditional maintain. The 45/25/25/5 portfolio (CSPX, Bitcoin, BBCA, gold) suits only aggressive investors with a 10+ year horizon, a separate cash buffer, and tolerance for paper losses of 30% or more. Run it with quarterly tax-efficient inflow steering and a 5 percentage point rebalancing band.

Metric (10.8 years, Jan 2016 – Oct 2026)	Result
CAGR, weekly calendar-aligned	+28.66%
Annualized volatility	21.74%
Sharpe ratio (Rf 4.0%)	1.13
Maximum drawdown, weekly close	-31.99%
Backtest ending wealth, nominal / real	$555,260 / $403,402
Money-weighted return (XIRR)	+27.31%
Benchmark CAGR (60/40 global equity-bond)	+8.61%

What drives the result.

- Bitcoin supplied about 66% of the active return versus the 60/40 benchmark, and the portfolio's correlation to BTC is 0.86 even though BTC is 25% of target weight.
- Rebalancing and compounding added about 4.41 percentage points a year in the frictionless model, roughly 22% of active return. Taxes and spreads will reduce this.
- BBCA returned +8.01% in IDR but +5.29% in USD, a 2.72 point currency drag.
- Diversification weakens under stress: average pairwise correlation rises from 0.084 in calm weeks to 0.341 in crisis weeks.

Read the history with care. The window is short, and Bitcoin's best calendar year (+1,424.72%) sits inside it. Past CAGR should not be extrapolated. The Monte Carlo and walk-forward modules were flagged by the engine as not recalibrated, so this memo excludes them from the findings (see the last section).

Portfolio and methodology

The portfolio is USD-based and holds four positions. Returns and risk use Friday-to-Friday weekly compounding (52 periods a year), which keeps Bitcoin's 24/7 weekend returns from being diluted by 5-day equity calendars.

Asset	Instrument	Weight	Quote currency	Role
CSPX	iShares Core S&P 500 UCITS ETF (Acc), CSPX.L	45%	USD	Core US equity
BTC	Bitcoin, BTC-USD	25%	USD	High-convexity growth
BANK	PT Bank Central Asia, BBCA.JK (net of 10% dividend tax)	25%	IDR	Indonesian financial equity
GOLD	SPDR Gold Shares, GLD	5%	USD	Partial hedge

Data quality. The audit found no duplicates. Forward-filling covered 28.79% to 32.68% of daily rows for BBCA, CSPX, GLD and USDIDR because of exchange-calendar gaps; BTC had none. It flagged 146 return outliers, many of which are genuine events such as March 2020. The risk-free rate is a constant 4.00%; actual 3-month T-bill yields averaged 0.93% over 2016–2021 and 2.27% over the full period, so Sharpe ratios are conservative relative to the early years.

Module	Confidence	Reason
Weekly risk and return	Medium-high	Calendar-aligned method
Correlation and regimes	Medium	Crisis subsample is only 141 weeks; selection bias applies
Backtest with contributions	Medium	Includes fees; may omit slippage, custody and exact tax timing
Tax engine	Medium	Sound structure, but legal assumptions need verification
Stress tests	Low-medium	Single-factor, deterministic shocks
Monte Carlo, walk-forward	Excluded	Flagged by the engine as not recalibrated

Performance and risk

The portfolio earned 28.66% a year with 21.74% volatility, with Bitcoin supplying most of the return and BBCA the weakest risk-adjusted result.

Asset	CAGR	Ann. volatility	Sharpe	Max drawdown	Weekly VaR 95%	Weekly CVaR 95%
CSPX	+15.24%	15.77%	0.693	-29.51%	2.93%	4.91%
BTC	+61.53%	66.68%	0.873	-81.69%	12.92%	18.51%
BANK (USD)	+5.56%	24.67%	0.052	-58.65%	4.50%	7.78%
GOLD	+12.36%	15.42%	0.560	-24.21%	3.13%	4.41%
Portfolio	**+28.66%**	**21.74%**	**1.130**	**-31.99%**	**3.79%**	**6.14%**

Other portfolio statistics: Sortino 1.742, Calmar 0.896, skewness -0.415, excess kurtosis 4.516 (fat tails), weekly VaR 99% 6.91% and CVaR 99% 10.79%. The best calendar year was +158.21% and the worst -24.73%. The drawdown of -31.99% exceeds the worst calendar year, so losses can build across year-ends.

BBCA is the clear laggard: a Sharpe of 0.052 and a -58.65% drawdown for a 25% weight.

Correlation and regime risk

Diversification is real in calm markets and thinner in stress. Weekly correlations are low across the four assets, but the portfolio itself tracks Bitcoin at 0.86.

Weekly correlation	CSPX	BTC	BANK	GOLD
CSPX	1.00	0.21	0.36	0.19
BTC	0.21	1.00	0.13	0.13
BANK	0.36	0.13	1.00	0.14
GOLD	0.19	0.13	0.14	1.00
Portfolio	0.60	0.86	0.50	0.24

The engine split weeks by rolling 4-week volatility of an equal-weighted basket: the top quartile (141 weeks) is the crisis regime and the other 424 weeks are normal.

Regime	Weeks	Average pairwise correlation
Normal	424	+0.084

Regime	Weeks	Average pairwise correlation
Crisis	141	+0.341
Shift		+0.257

The largest jump is CSPX–BBCA, from 0.21 to 0.52. Gold's correlations also rise, to 0.29–0.38. Conditioning on high volatility mechanically inflates correlations (Forbes and Rigobon, 2002), so this shows higher co-movement during shocks, not proof of contagion. The practical point stands: diversification is weakest when it is needed most.

Backtest and rebalancing policy

The base case invests $10,000 plus $107,296 of monthly contributions ($117,296 total) and ends at $555,260 nominal, or $403,402 in today's purchasing power: 4.73x nominal and 3.44x real. Net gains after $720.89 of fees are $437,963. The time-weighted CAGR is +31.52% (+27.69% real) and the XIRR is +27.31%; XIRR is the most investor-relevant figure because it reflects contribution timing. The backtest CAGR is higher than the +28.66% weekly figure because it uses quarterly rebalancing with contributions rather than the daily-rebalanced portfolio.

Strategy	Ending nominal	TWR CAGR	XIRR	Max drawdown	Sales	Fees
Quarterly calendar (standard)	$561,743	+31.75%	+27.50%	-35.29%	43	$749.86
Threshold 5% band (standard)	$462,683	+28.68%	+24.30%	-33.93%	61	$590.51
Tax-efficient inflow, quarterly steering	**$555,260**	**+31.52%**	**+27.31%**	**-35.30%**	**43**	**$720.89**
Tax-efficient inflow, 10% corridor	$498,743	+29.92%	+25.54%	-35.22%	18	$426.08
Inflow-only, zero sales	$777,764	+36.52%	+32.82%	-65.46%	0	$117.30

Policy: quarterly inflow steering with a 5 point band. New cash goes first to the most underweight asset; sell only if a position still exceeds target by more than 5 points. Zero-sell produced the most wealth but a -65.46% drawdown consistent with unchecked concentration drift, which most investors would not hold through. Standard quarterly calendar rebalancing ended slightly higher than steering ($561,743 vs $555,260); the gap is small, and the engine rates steering as the tax-optimized option.

Allocation alternatives (static weekly rebalancing, so not directly comparable to the table above):

Allocation	Ann. return	Volatility	Sharpe	Max drawdown
Fixed 45/25/25/5	+27.99%	21.93%	1.094	-32.45%
Equal weight	+27.69%	21.14%	1.121	-30.22%
Risk parity (BTC 8.6%, gold 37.3%)	+18.13%	13.76%	1.027	-24.41%
Minimum variance (BTC 0%, gold 48.7%)	+13.68%	11.84%	0.817	-20.74%

Risk parity kept a Sharpe of 1.027 with about 8 points less drawdown. An investor unable to hold -30% to -35% should cut Bitcoin toward 10–15% rather than abandon rebalancing discipline.

Tax review

Using Ireland-domiciled, accumulating CSPX instead of a US-domiciled S&P 500 ETF such as VOO adds about 22.5 basis points a year for a non-US resident investor in the model. The modeled gain is small early and compounds with time.

Horizon	CSPX net wealth	VOO net wealth	Tax alpha	Boost
5 years	$28,782	$28,577	+$206	+0.72%
10 years	$80,683	$79,678	+$1,006	+1.26%
20 years	$384,642	$375,968	+$8,674	+2.31%
30 years	$1,443,308	$1,396,660	+$46,648	+3.34%

The sources are the 15% US–Ireland treaty rate at fund level versus the 30% US statutory rate for non-resident aliens, the accumulating structure (no cash distribution drag), and no US estate tax exposure through the Irish fund.

To verify before client use.

- GLD is treated as 0% withholding on an assumption, not a verified fact. As a US-domiciled fund it may carry estate-tax and reporting exposure for non-US residents, so confirm treatment or consider a non-US gold vehicle.
- BTC (0%) and BBCA (10% dividend withholding) rest on Indonesian rules the engine marks as verified; confirm the client's residency and treaty status and keep cost-basis records.

Stress tests

A 2008-style equity crash is the most damaging scenario at -33.4%, because CSPX, BTC and BBCA all fall together. Gold at 5% offsets little in any case. These are deterministic single-scenario shocks applied to $100,000, not probabilities.

Scenario	CSPX	BTC	BANK	GOLD	Portfolio	Post-shock value
Equity crash 2008	-35%	-45%	-28%	+12%	-33.40%	$66,600
Crypto winter	-5%	-65%	-2%	+4%	-18.80%	$81,200
EM currency crisis, USD spike	-8%	-15%	-35%	+15%	-15.35%	$84,650
Stagflation	-22%	-25%	-15%	+28%	-18.50%	$81,500

Even when gold rises 28% in stagflation, the portfolio still falls 18.5%. A joint-shock model with correlated moves would be a better next step than these single-factor scenarios.

Benchmark and attribution

The portfolio beat a 60/40 global equity-bond benchmark by 20.05 points a year, but the comparison is not like for like: the benchmark holds no Bitcoin, no Indonesian single stock and no gold, so the result measures allocation risk, not manager skill.

Metric	Portfolio	60/40 benchmark
CAGR	+28.66%	+8.61%
Volatility	21.74%	10.45%
Sharpe (Rf 4%)	1.134	0.441
Max drawdown	-31.99%	-22.06%
Tracking error	17.96%	
Information ratio	1.117	
Beta / correlation to benchmark	1.189 / 0.571	

Source of active return (+20.05 pts)	Active contribution	Share
BTC	+13.23	66.0%
Rebalancing and compounding	+4.41	22.0%
CSPX	+2.98	14.9%

Source of active return (+20.05 pts)	Active contribution	Share
GOLD	+0.19	0.9%
BANK	-0.76	-3.8%

This is a simplified weight-times-excess-CAGR attribution, not a full Brinson analysis, because the benchmark has no equivalent for BTC, gold or Indonesian equities. A custom policy benchmark (45% S&P 500, 25% BTC, 25% BBCA net total return in USD, 5% gold) would be a fairer yardstick.

Interest-rate sensitivity. Regressing weekly returns on 10-year Treasury yield changes (564 weeks), the portfolio has no significant duration exposure (beta +0.0004, p = 0.97). The benchmark shows an implied duration of 1.97 years, and a 100% BND sanity check gives 5.47 years with R^2 of 0.79, in line with the roughly 6-year duration of the US aggregate index. Rate risk in the portfolio is therefore indirect, through equity valuations and risk appetite.

Risks, monitoring and limitations

The main risks are Bitcoin concentration, a 25% single-stock position, and weak diversification in crises. The escalation triggers below are proposed analyst thresholds, not outputs of the engine.

Risk	Evidence	Mitigation
Bitcoin concentration	66% of active return; 0.86 correlation to portfolio	5-point band; consider 10–15% BTC for lower-risk clients
BBCA single-stock and FX risk	Sharpe 0.052; -2.72 pt FX drag	Review concentration; confirm client base currency
Crisis correlation	Average pairwise 0.084 to 0.341	Hold a separate cash buffer; no leverage
Tax assumptions	GLD treatment unverified	Written tax position before client use
Regime dependence	One 10.8-year window	Do not extrapolate historical CAGR

Proposed trigger	Threshold
BTC weight	Above 35%

Proposed trigger	Threshold
Portfolio drawdown from peak	Beyond -25% to -30%
Average pairwise correlation	Above 0.40
Suitability review	Horizon under 10 years or near-term liquidity need

Limitations. Results are historical and net of modeled fees only; they exclude slippage, custody costs and client-specific taxes. Forward-filling affects roughly 30% of daily rows for several series. The Monte Carlo and walk-forward modules still use raw daily data (the engine labels them internal diagnostics) and are excluded from this memo; they should be rebuilt on weekly returns, with taxes, fees, USD/IDR and a contribution schedule, before any forward-looking figures are shown to clients. This memo is analysis, not individualized investment advice.